Problem Set **#1**

P1. Answer with reason. If the intrinsic carrier concentration (n_i) in Si, Ge and GaAs samples at room temperature are compared, which of the following statements is true.

(a) n_i of Si > n_i of Ge > n_i of GaAs
(b) n_i of GaAs > n_i of Si > $n_{i\,i}$ of Ge
(c) n_i of Ge > n_i of Si > n_i of GaAs

P2. Island-X is very special. Here only twins are born, one male and the other female. Also, on this island, partners (a male and a female) die simultaneously. On this island, 1,210 pairs of twins are born every year. The number of couples dying per year is 0.001% of the product of the male population (m) and female population (f). (The logic is as follows: The number of deaths is proportional to the male population. It is also proportional to the female population. Hence, the number of deaths is proportional to the product of the male and female populations.)

(a) If the population of the island was stable over many years, what are the values of m and f?

(b) The people of the island decided to introduce birth control measures and brought down the birth rate to 1,000 pairs of twins/year. After some years, the population of the island stabilized. What are the new values of m and f?

(c) Relate the above to generation, recombination and intrinsic carrier concentration in a semiconductor.

(d) Then suddenly in one year, 1,000 males invaded the island. They slowly integrated with the other inhabitants of the island. When the population stabilized after some years, what are the new values of m and f? Note that the birth rate is still 1,000 pairs of twins/year.

(e) If 10,000 males had invaded the island, what would have been the new values of m and f after the population stabilized?

(f) What is the product of m and f in (b), (d) and (e)? What can you conclude?

(g) If, after an invasion, the value of m had stabilized at 12,500, what would be f and the number of invaders?

(h) Relate (d) – (f) to the effect of doping in semiconductors.